



Improving access to emergency medical services using advanced air mobility vehicles

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Accepted: 2 August 2023 / Published online: 18 September 2023
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Abstract

The latest advancements in electric vertical take-off and landing (eVTOL) vehicles indicate that soon this technology will be available in multiple fields. One potential application of this new technology is in emergency medical services. These vehicles will be able to reach emergency sites faster than ground ambulances at lower costs than traditional helicopters. So in the following years, eVTOL vehicles could be used for aeromedical transportation. One crucial decision in implementing such a technology in emergency medical services is the location of their take-off and landing areas (vertiports). In this work, we propose a methodology for locating the vertiports in a healthcare network to improve emergency medical services coverage in hard-to-reach zones. We studied the system performance locating the vertiports for emergency services in existing healthcare facilities or outside them as auxiliary bases. In addition, we evaluated the performance of different operational scenarios regarding the use of emergency eVTOL. To do so, we used data analytics techniques (i.e., clustering algorithms) in conjunction with facility location models. The approach is tested using data from the Auvergne-Rhône-Alpes region in France. Results showed that locating the vertiports in existing healthcare facilities is the best choice in terms of coverage of hard-to-reach zones. However, on average, the response times increased compared to locating the vertiports as auxiliary bases outside the healthcare facilities. Besides, the results indicated that implementing eVTOL vehicles for aeromedical transportation can provide better access to emergency medical services in hard-to-reach zones. Still, the autonomy of such vehicles plays an essential role in their applicability.

Keywords Emergency medical services · Advanced air mobility · Healthcare · Clustering · Facility location

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1 Introduction

Assisting patients during an emergency is a vital service in all countries worldwide. To do so, the healthcare providers count on paramedic teams who can be deployed quickly to the emergency sites. In most cases, the paramedics reach the patient's location using emergency vehicles such as ambulances, helicopters, or rapid response vehicles. A critical factor in emergency medical services (EMS) is the response time. Longer response times are related to worse outcomes, especially in trauma patients (Mell et al. 2017). Thus health professionals are expected to arrive at the area of the emergency as soon as possible. The delay in the response depends on multiple factors, including location and type of emergency vehicles used to serve the emergency (Reuter-Oppermann et al. 2017).

In recent years, there have been considerable advances in electric vertical take-off and landing (eVTOL) vehicles. It is suggested that soon these vehicles could safely and efficiently transport people and cargo in urban and rural areas (Cohen et al. 2021). Nowadays, multiple projects are aimed at developing and expanding new air services worldwide. Examples of these initiatives are the CityAirbus NextGen, an eVTOL with four seats (Airbus 2021); or the Uber and Hyundai S-A1, an eVTOL that will be able to transport four passengers (Hyundai 2020). Besides, NASA's Advanced Air Mobility Mission is focused on developing air transportation systems for emerging aviation markets that use eVTOL vehicles (NASA 2022). In addition, Boeing and Wisk published their proposed Concept of Operations for uncrewed passenger-carrying in urban air mobility environments (Boeing 2022). According to Thipphavong et al. (2018), apart from package delivery or passenger transportation, new air transportation services could include emergency services, humanitarian missions, etc. The previous innovative air services comprise the advanced air mobility (AAM) paradigm, which focuses on emerging aviation markets for on demand aviation services between urban, suburban, and rural areas (Goyal et al. 2021).

This work explores the potential use of AAM aircraft in EMS. In our opinion, healthcare services such as the EMS can be improved using this new technology. Currently, some health providers have helicopters that serve as air ambulances. These vehicles have the advantage of traveling faster than traditional ground ambulances, which in some cases can be the difference between life and death. However, the cost of providing air EMS using helicopters is high. Indeed, a report from the U.S. Government Accountability Office (2017) showed that the average cost per flight from eight providers ranged between 6,000–13,000 USD in 2016. Furthermore, the prices charged to patients for receiving this service can reach tens of thousands of dollars due to the lack of regulation of this service (Kelly 2020). Besides, according to the Association for Air Medical Services (2017), approximately 550,000 EMS requests are served using helicopters every year in the U.S. That is less than 2% of the estimated 36 million EMS transport services made in the U.S. in a year (National Emergency and Medical Services Information System (NEMSIS) 2019).

On the contrary, the costs of future AAM aircraft seem to be lower. Garrow et al. (2021) reported that Uber forecasts the flying costs per hour of AAM

vehicles at US 662 USD/h, which is lower than the 1,253 USD/h of the helicopters today. We believe that AAM aircraft used for emergency services can offer the advantages of the traditional emergency aeromedical transportation (using helicopters) at lower costs and with additional operational advantages. In addition, the study of Goyal and Cohen (2022) concluded that with certain technological improvements in AAM, aeromedical transport made by eVTOL aircraft could be more cost-effective and reliable.

In this work, we studied how the location of vertiports for AAM can improve access to EMS in hard-to-reach zones. These are in rural or remote areas with limited road infrastructure, which makes very difficult to access them by ground. EMS provided by air services can improve access to healthcare services to unserved communities that travel significant distances to receive health services (Chappelle et al. 2018). By using data analytics techniques (i.e., clustering), facility location models, and considering the eVTOL aircraft characteristics, we determine the location of vertiports with the objective of improving the EMS coverage in hard-to-reach zones. Our work evaluated EMS system performance when i) locating the vertiports in existing healthcare facilities with an emergency department, and ii) locating them outside as auxiliary bases. In addition, we analyzed and compared different operational scenarios in terms of coverage, response time, and operational service time. In this work we refer to healthcare facility as the hospital or clinic with an available emergency department.

To prove the practicality of our approach, we used data collected from the Auvergne-Rhône-Alpes region in France. This data relates to the healthcare network, potential locations where medical emergencies can occur, and EMS response time objective for rural areas. Similar studies have been conducted to study the use of AAM aircraft to improve other services. For instance, Rajendran and Zack (2019) analyzed data from taxi rides in New York City, USA using an iterative clustering approach proposed for air taxi vertiport location. However, to the best of our knowledge, no other studies have explored the use of AAM technology in EMS considering facility location decisions.

Based on our results, the contributions of this work can be summarized as follows:

- We studied a relevant application of AAM in healthcare services.
- We introduced a methodology that identifies the hard-to-reach zones for EMS and make facility location decisions for vertiports.
- We analyzed and compared EMS system performance when locating the vertiports in existing healthcare facilities with emergency departments and locating them as auxiliary bases in different locations.
- We validated the methodology using real data from a case study from the Auvergne-Rhône-Alpes region, France.

The main results indicated that the introduction of AAM technology can be beneficial to increase EMS accessibility to hard-to-reach zones compared to the current condition. However, implementing this technology is highly dependent on advancing this type of technology, specifically in flying autonomy. In addition, the main

results indicated that in terms of coverage, the best option is to locate the vertiports in existing healthcare facilities with emergency departments. However, in terms of response time, locating the vertiports as auxiliary bases is the best choice.

The remainder of the paper is organized as follows: Sect. 2 reviews the related works; Sect. 3 presents the solution approach used in this work; Sect. 4 describes the case study at the Auvergne-Rhône-Alpes region, France and details the data collection section and the assumptions made in this work; Sect. 5 analyzes the results of the experiments; and Sect. 6 provides the conclusions of this work.

2 Related works

In recent years, there have been multiple experiences with using AAM technologies in the healthcare sector worldwide. One early use case was the transportation of laboratory samples for early infant HIV diagnosis using drones in Malawi, Africa made by UNICEF. Also, this NGO worked with the Malawi government to participate in emergency flood responses with nonconventional aircraft (UNICEF 2018). Another early use case was the drone-based delivery system for blood and urgent medical supplies transportation established by the company Zipline in Rwanda, Africa (Ackerman and Koziol 2019). More recently, with the emergence of the COVID-19 pandemic, some medicines, medical supplies, and tests were delivered to remote areas using AAM aircraft. For instance in the U.S., the company CSV Pharmacy partnered with UPS to deliver prescription medicines to a retirement community in Florida (Premack 2020). In addition, during the COVID-19 pandemic, Zipline established a drone-based drug refill delivery system in remote areas where patients were not able to attend oncology centers (Umutesi et al. 2021). Currently, drones are being used to deliver health supplies in countries such as Scotland, Canada, the Democratic Republic of the Congo, the Republic of Vanuatu, etc. (Snouffer 2022). These are some examples of how AAM technologies, especially drones, have increased access to healthcare services in remote and unserved areas.

Moreover, research on AAM technologies applied to healthcare services have mainly focused on the use of drones. For instance, Kim et al. (2017) studied the drone-aided delivery and pickup planning of medication and test kits for patients with chronic diseases. Asadi et al. (2022) addressed the distribution operations of a drone swap station with the objective of maximizing customer demand fulfilled using drone flights. Al-Rabiaah et al. (2022) studied the problem of selecting the locations of drone launching centers while maximizing patient coverage within drone range constraints. However, providing EMS using AAM technology implicates aeromedical transportation using eVTOL vehicles. Research on this subject is scarce. Among the few works, the exploratory study of Chappelle et al. (2018) concluded that near-term eVTOL aircraft is not expected to fulfill the performance requirements for aeromedical transportation in terms of range. In addition, the authors concluded that advancements in eVTOL autonomy and sense and avoid technologies may be the key to increasing safety and reducing the costs of aeromedical transportation provided by AAM aircraft. Lastly, this study concluded that new operational models where eVTOL aircraft perform secondary missions could

bring financial viability to this AAM application. Mihara et al. (2021) presented a cost analysis of eVTOL aircraft for EMS purposes in Japan. Among the different conclusions drawn in this study, the authors pointed out that the difference in battery capacity and its high cost of replacement would significantly impact future cost models. A recent study of Goyal and Cohen (2022) concluded that to increase the reliability and cost efficiency of aeromedical transport using eVTOL aircraft, charging times have to be reduced, operational ranges have to be increased, and effective battery strategies have to be implemented.

In recent years, there has been growing interest in vertiport location decisions, also known as the vertiport selection problem. One of the earliest studies addressing this problem is the work of Rajendran and Zack (2019). These authors proposed a methodology for locating vertiport and vertistop locations for air taxis in New York City. To do so, they used a multicriteria warm start technique in conjunction with a k-means clustering algorithm. Apart from proposing 18 site locations, the authors also concluded that the most important factors in vertiport selection decisions for air taxis are the on-road travel limit and percentage of customer demand satisfaction. Willey and Salmon (2021) formulated the vertiport selection problem using a single allocation p-Hub median location problem. They included AAM features like vehicle speed, battery range, desired operational strategies, and the possibility of trips involving more than two vertiports. As solution methodologies, they presented five heuristic algorithms where the most effective on average is within 10% of the optimal solution. Another relevant work on the vertiport selection problem is the study of Jeong et al. (2021), who proposed a methodology for establishing vertiport locations contemplating the effect of aircraft noise in residential zones of the Seoul metropolitan area. First the authors analyzed and clustered demand data from the commuter population identifying potential vertiport locations over the region under study. Next they selected the final vertiport locations under two scenarios: the first one considers the shortest route between commuters' origin to destination (business scenario); and the second scenario selects the vertiport location minimizing the amount of noise exposure in residential areas. As a result of the case study, the noise priority scenario led to a reduction of 76.9% in the population affected by high levels of noise compared to the business priority scenario. More recently, Chen et al. (2022) presented an uncapacitated single allocation p-Hub median model for locating vertiports in a metropolitan area minimizing the total travel cost. The authors modeled the region using a grid structure, considering the possibility of excluding some grid cells of the set of candidate vertiport locations. A variable neighborhood search heuristic is proposed that was able to optimally solve most of the instances. Another recent contribution on the vertiport selection problem is the work of Shin et al. (2022). In this study, the authors presented a hub location model which minimizes the total cost comprised of facility costs, travel costs and collision risk costs. The authors proposed a heuristic algorithm based on a genetic algorithm to solve larger instances. Numerical results suggest that the introduction of air taxi services requires careful consideration, especially when the risk of aircraft midair collision is high.

Based on the literature review presented, only some studies have addressed the use of AAM technology in EMS. These works have focused on cost analysis and

examining the potential market and operational viability of AAM technology in EMS. It has yet to be studied facility location decisions considering operational scenarios for emergency eVTOL aircraft. Our work proposes a solution procedure for the vertiport selection problem in EMS that address this literature gap.

3 Solution approach

In this section we present the methodologies used for making vertiport location decisions (see Fig. 1). We first present the methodology for locating the vertiports in existing healthcare facilities with emergency departments; and next for establishing vertiports in new locations or bases.

3.1 Locating vertiports for EMS in existing healthcare facilities with emergency departments

The first step is to collect and preprocess data regarding the potential emergency sites and the healthcare network. Next we determine the hard-to-reach zones where certain criteria are met in terms of response times and time-savings. Lastly, the facility location decisions are made considering the hard-to-reach zones and the vertiport candidate locations. The detail of each step is presented next.

3.1.1 Data preparation and clustering

First we determine the sites where medical emergencies can occur in the region under study. Since medical emergencies can occur almost everywhere, the number of potential emergency sites is large and can reach millions of data points. It includes

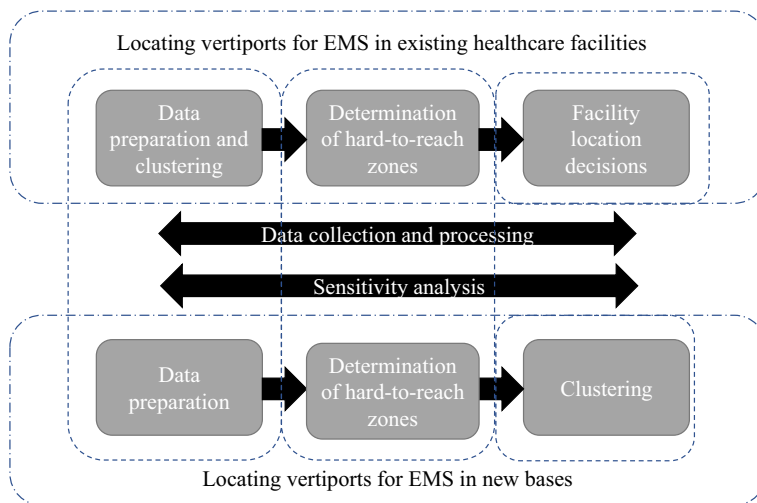


Fig. 1 Methodological framework

households, workplaces, commercial sites, roads, train stations, rails, parks, etc. Making facility location decisions considering the large data under consideration is computationally expensive and unsuitable for this study. Therefore, we use the well known K-means algorithm presented in Bock (2008) to reduce the number of points and group them into K clusters of potential emergency sites. Multiple works such as by Kim and Ham (2019) or Wang et al. (2020) have used clustering algorithms to aggregate geographical coordinates. In this work, we used the K-means algorithm presented in the work of Manning et al. (2009). Notably, it is possible to know how many data points or sites are grouped in each cluster. This information will be used in the facility location model of this proposal.

3.1.2 Determination of hard-to-reach zones

The next step is to determine the subset of clusters that would benefit the most from EMS provided by AAM technology. We refer to these clusters as hard-to-reach zones or target clusters that meet at least one of the following criteria. The first criterion is if the maximum response time δ is not met. It means that ambulances are not reaching the sites of emergencies within an established amount of time. The second criterion is that the time-savings for reaching emergency sites using eVTOLs is at least the percentage λ of the traveling time of a ground ambulance. We define time-savings as the time saved in an emergency response as a result of using an emergency eVTOL in comparison to a ground ambulance. In some cases, arriving earlier to emergency sites can improve the patient's chance of survival. As an example, if an emergency eVTOL aircraft can reach a potential emergency cluster in half the time of a ground ambulance and λ is set to 50% or less, this cluster meets this criterion and is classified as hard-to-reach zone. However, if λ is set to more than 50% this criteria is not met. We determine the hard-to-reach zones by adapting the procedure of Rajendran and Zack (2019), which was proposed for air-taxi ride eligibility as follows.

Consider a set of population clusters K situated over a region. Each cluster $c \in K$ has a centroid in the latitude Lat_i and the longitude Lon_i . For providing EMS with AAM vehicles, a set of healthcare facilities with emergency departments has the possibility of establishing vertiports. Each medical facility $m \in M$ is in the latitude Lat_m and the longitude Lon_m . Due to the urgency of each EMS request, it must be served in the nearest medical facility. Thus, for each cluster $c \in K$ we identify the nearest medical facility providing EMS. We refer to the latitude and longitude of the nearest medical facility of each cluster $c \in K$ as $LatN_c$ and $LonN_c$, respectively. Next we calculate the air travel time $airt_c$ between each cluster c and its nearest medical facility. The air travel time is comprised by a preparation time and the on-route time. The preparation time α corresponds to the time between receiving the EMS request and the emergency eVTOL take-off. The on-route time is the time between the eVTOL take-off and the arrival to the emergency cluster. We calculate the on-route time considering the eVTOL average speed ($speed_{evtol}$) and the distance to the nearest medical facility. To calculate the actual distance flown by an eVTOL aircraft we used the Haversine formula denoted in this work as d_{ij} and detailed in Appendix 1. The Haversine formula has been used for computing the flying distance traveled by

various types of aircraft including emergency helicopters (Tassone et al. 2020) and eVTOL aircraft (Lim and Hwang 2019). Equation 1 summarizes the $airt_c$ calculation for each cluster $c \in K$.

$$airt_c = \alpha + \frac{d_{ij}(Lat_c, Lon_c, LatN_c, LonN_c)}{speed_{evtol}} \quad (1)$$

Next for each cluster $c \in K$, Eq. 2 calculates the time-savings for using the AAM aircraft compared with a ground ambulance. We denote the time-savings in the parameter $timesav_c$ and the ambulance travel time in the parameter $ambt_c$.

$$timesav_c = \max(0, airt_c - ambt_c) \quad (2)$$

Lastly, Eq. 3 evaluates if the clusters meet the criteria for being classified as hard-to-reach zone. The first part of the equation observes the maximum response time δ , and the second part of the equation observes the λ time-savings condition. If a cluster meets at least one criterion, it is classified in the target cluster subset (hard-to-reach zones) that is used in the facility location model of the next subsection.

$$E_c = 1 \rightarrow (if \ ambt_c > \delta) \vee (ambt_c \lambda < timesav_c); 0 \rightarrow otherwise \ \forall c \in C \quad (3)$$

3.1.3 Maximal covering location problem

Once we determine the hard-to-reach zones represented by the target clusters, the next step is to determine the medical facilities that, by establishing a vertiport, would increase the EMS coverage in these zones. To do so, we use the maximal covering location model. The covering problems are appropriate for determining the location of EMS facilities (Ahmadi-Javid et al. 2017). Since we have a set of candidate medical facilities in which the vertiports for EMS will be established, this model corresponds to a discrete facility location problem. We adapted the maximal covering location model presented in Owen and Daskin (1998) as follows.

Consider a set J of potential medical facilities with emergency departments with the possibility of establishing a vertiport for supporting its EMS department. In addition, the set I represents the target clusters to be covered (determined in Subsect. 3.1.2). Each target cluster $i \in I$ has a demand h_i that represents the number of sites grouped in the cluster (determined in Subsect. 3.1.1). The binary parameter a_{ij} is 1 if the cluster $i \in I$ is within the range $\emptyset$ of an eVTOL located in a vertiport candidate location $j \in J$. Lastly, the parameter P denotes the number of candidate locations to be established. Notably, the parameter a_{ij} also depends on how the eVTOL aircraft are used for attending to emergency requests. For example, a patient can either be transported back to the healthcare facility where the emergency eVTOL departed or be transported to the nearest emergency department resulting in different operational scenarios (OS). Figure 2 presents the OS addressed in this work when locating vertiports in existing emergency departments.

The path of the emergency eVTOL for the OS under study (Fig. 2) are the following:

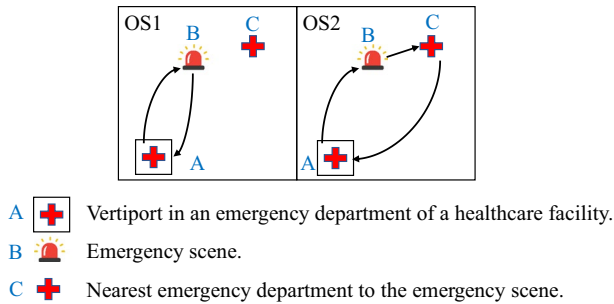


Fig. 2 OS for EMS with vertiports located in existing healthcare facilities with emergency department

1. Traveling to the emergency site (A-B) and transporting the patient back to healthcare facility from where the emergency eVTOL departed (B-A).
2. Traveling to the emergency site (A-B), transporting the patient to the nearest emergency department (B-C), and returning to the healthcare facility from where the emergency eVTOL departed (C-A).

Thus, for calculating the parameter a_{ij} for OS1, we consider two times (roundtrip) the distance between the vertiport candidate location $j \in J$ and the emergency cluster $i \in I$ as follows.

$$a_{ij} = \begin{cases} 1, & \text{if } 2 \times d_{ij}(Lat_i, Lon_i, Lat_j, Lon_j) \leq \emptyset. \\ 0, & \text{otherwise.} \end{cases} \quad (4)$$

For OS2, the a_{ij} parameter is computed calculating the distance between (i) the vertiport candidate location $j \in J$ and the emergency cluster $i \in I$; (ii) the emergency site $i \in I$ and its nearest emergency department located in $(LatN_i, LonN_i)$; and (iii) returning to the vertiport candidate location $j \in J$.

$$a_{ij} = \begin{cases} 1, & \text{if } d_{ij}(Lat_i, Lon_i, Lat_j, Lon_j) + d_{ij}(Lat_i, Lon_i, LatN_i, LonN_i) + d_{ij}(LatN_i, LonN_i, Lat_j, Lon_j) \leq \emptyset \\ 0, & \text{Otherwise} \end{cases} \quad (5)$$

We used the following decision variables for the facility location model.

$$\begin{aligned}
 x_j &= \begin{cases} 1, & \text{if vertiport is located at healthcare facility } j \in J. \\ 0, & \text{otherwise.} \end{cases} \\
 y_{ij} &= \begin{cases} 1, & \text{if target cluster } i \in I \text{ is covered by the candidate location } j \in J. \\ 0, & \text{otherwise.} \end{cases}
 \end{aligned}$$

Using the previous definitions, the facility location model can be formulated as follows.

$$Max \sum_{i \in I} \sum_{j \in J} h_i y_{ij} \quad (6)$$

Subject to:

$$\sum_{i \in I} y_{ij} \leq 1 \quad \forall j \in J \quad (7)$$

$$y_{ij} \leq a_{ij}x_i \quad \forall i \in I; \forall j \in J \quad (8)$$

$$\sum_{j \in J} x_j \leq P \quad (9)$$

$$x_j \in \{0, 1\} \quad \forall j \in J \quad (10)$$

$$y_{ij} \in \{0, 1\} \quad \forall i \in I; \forall j \in J \quad (11)$$

The objective function of Eq. 6 maximizes the EMS coverage to hard-to-reach zones. Constraint set 7 establishes that each target cluster is covered by one vertiport. Constraint set 8 observes the coverage range of the emergency eVTOL aircraft. Constraint set 9 limits the number of vertiports to P and last, constraints set 10 and 11 are the domain of the decision variables. The flowchart in Fig. 3 summarizes the logic of the methodology.

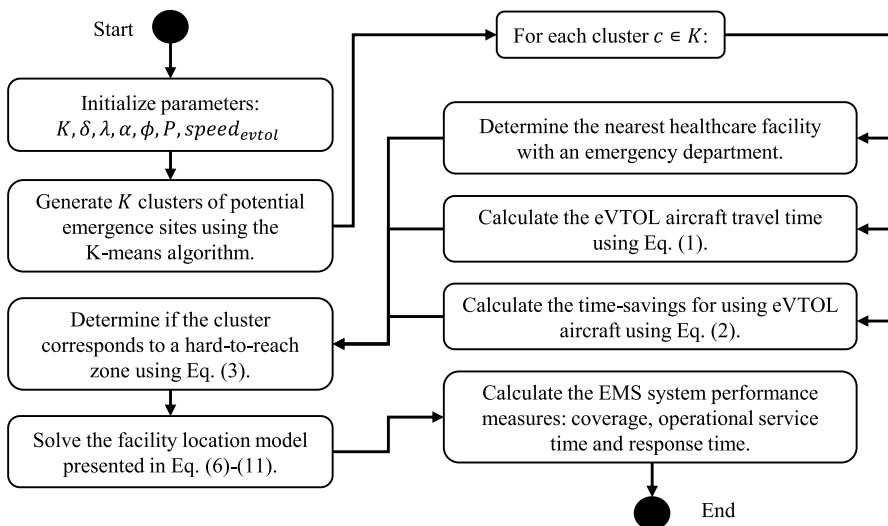


Fig. 3 Flowchart of the methodology for locating the vertiports in existing healthcare facilities

3.2 Locating vertiports for EMS in new facilities

Another possibility for locating vertiports for EMS in a healthcare network is to establish them as bases outside the healthcare facilities. In this way, the emergency eVTOL aircraft could be near to the hard-to-reach zones. However, this option requires a different OS which is presented in Fig. 4.

This OS is similar to OS2 of the previous subsection. However, in this case, the emergency eVTOL departs from an auxiliary base, travels to the emergency site (A-B), transports the patient to the nearest emergency department (C-A), and returns to the vertiport. To determine the location of the auxiliary bases we coupled a data analysis procedure with a clustering algorithm.

The outline of this solution procedure is the following. First we collect the information of the potential emergency sites and the healthcare network as presented in the first part of Subsect. 3.1.1. Next we determine the subset of potential emergency sites that are classified as hard-to-reach zones. The criteria to make this classification

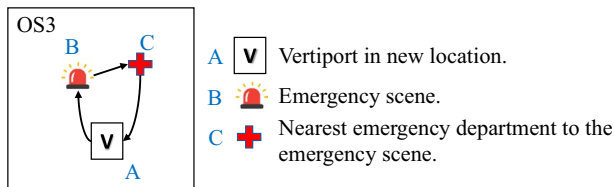


Fig. 4 OS for EMS with vertiports located in new facilities

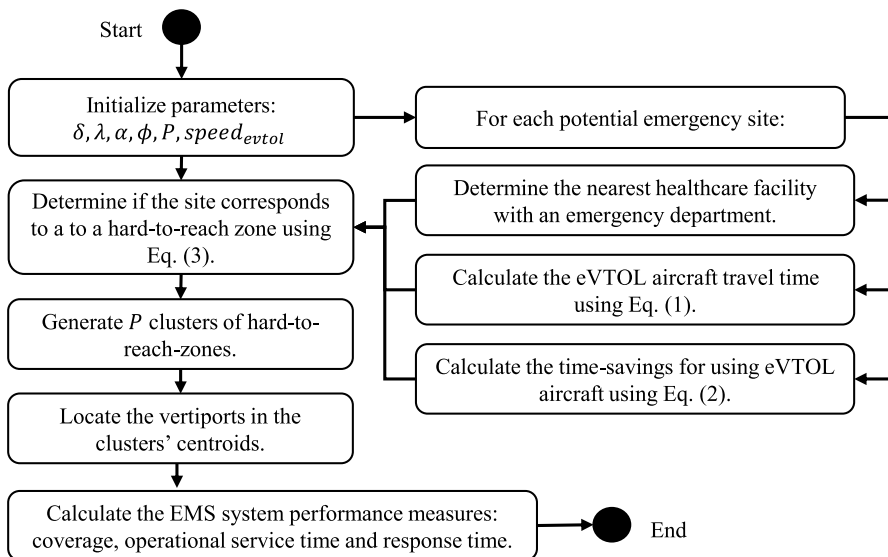


Fig. 5 Flowchart of the methodology for locating the vertiports outside the healthcare facilities

is that ground ambulances do not arrive in δ time units or the time-savings for using an emergency eVTOL is at least λ (Subsect. 3.1.2). Lastly, we use the K-means algorithm to cluster the hard-to-reach zones in groups where the vertiports for EMS will be located. The flowchart in Fig. 5 summarizes the logic of the methodology.

4 Case study description and data collection

The testing ground of this work was the Auvergne-Rhône-Alpes region in France. This is the second most populated region of France after the Île-de-France region. In 2019, the region had a population of 8,042,936 people and a density of 115.4 persons per square kilometer (Institut national de la statistique et des études économiques 2022). A big share of its population is concentrated in urban areas, such as the metropolis of Lyon, metropolis of Clermont Auvergne, and the cities of Saint-Étienne, Grenoble, and Annecy. Notably the urban areas registered the largest demographic growth over the last years as seen in Fig. 6, which also shows that an important share of the population lives in small villages or countryside areas of the region. EMS must be available to all persons regardless of the location they live.

In France, the SAMU (Service d'Aide Médicale Urgente) organization provides the EMS in the country. Each department has a SAMU unit that answers the local request for EMS to determine a proper response based on the nature of the request. If necessary, they can dispatch the SMUR (Service Mobile d'Urgence et

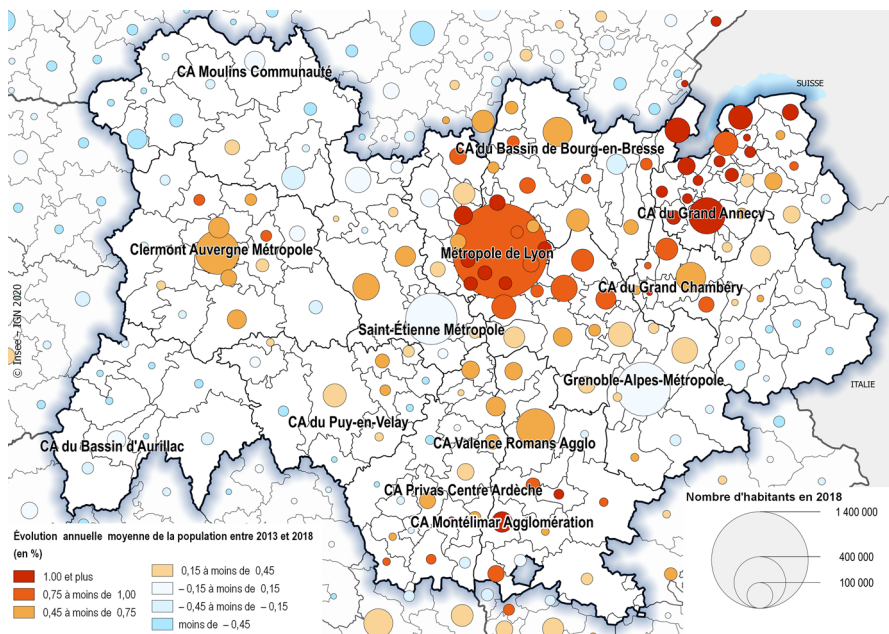


Fig. 6 Population density and demographic growth in the Auvergne-Rhône-Alpes region. Source: (Institut national de la statistique et des études économiques 2020)

Réanimation - Mobile Emergency and Resuscitation Service) units that count on rapid response vehicles, ambulances, and in some cases helicopters for reaching the emergency sites (SAMU Urgences de France 2022a). There are 48 SMUR units in the Auvergne-Rhône-Alpes region (SAMU Urgences de France 2022b). A request for EMS usually comes from a call from the site of the urgency. These calls are answered by the local SAMU authorities who determine the proper response based on an evaluation of the situation. If the situation requests the deployment of a SMUR unit, it will decide which vehicle will be used to reach the site of the emergency. It is worth noting that in France, the threshold for arriving at emergency sites is 30 min (Ministère des Solidarités et de la Santé 2015). In this case study, the healthcare facilities with emergency sites correspond to the hospitals with emergency departments or SMUR units.

4.1 Data collection

We obtained the latitude and longitude of the sites of the region under study from the data.gouv.fr portal, which is extracted from (OpenStreetMap 2021). These data consist of multiple cartographic layers of households, working places, commerce sites, roads, railways, etc., for each department in the region. We assumed that these cartographic layers correspond to the potential emergency sites in the region. Using the QGIS™ application, we established the centroids of the layers of interest and exported them. In total, we had 5,495,903 pairs of coordinates. After a cleaning process that corrected the erroneous (coordinates outside of the department) and duplicated data, we ended up with 5,495,574 pairs of coordinates. Next we collected and marked the locations of the SMUR units in the region. Figure 7 depicts the localization of the potential emergency sites and the SMUR units in the Auvergne-Rhône-Alpes region. It is worth noting that in the map of Fig. 7, there are 46 SMUR

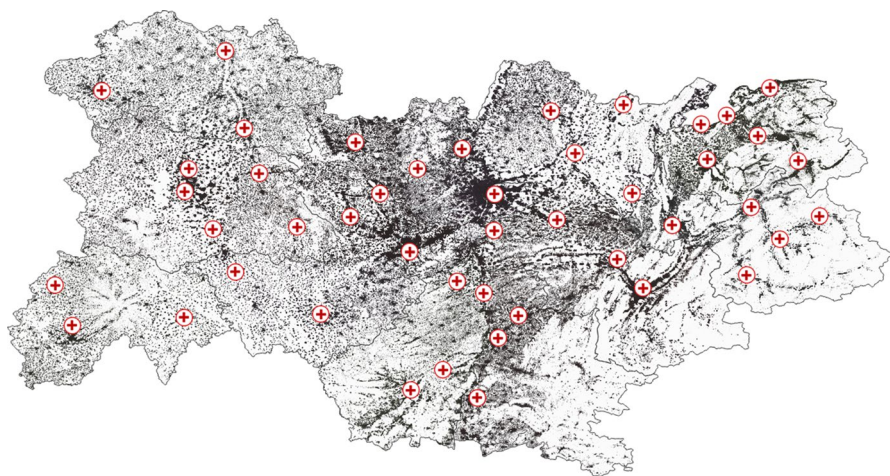


Fig. 7 Locations and the SMUR units in the Auvergne-Rhône-Alpes region, adapted from: (OpenStreet-Map 2021; SAMU Urgences de France 2022b)

locations marked rather than 48. This is because the SAMU 69, divided into the EST, NORD and SUD units, functions at the hospital Edouard Herriot in Lyon.

To obtain the ground ambulance travel distances and times between the sites and the emergency departments and to determine the nearest medical facility from the emergency clusters or sites, we used the Google Maps Distance Matrix API. It provides the travel time and distance for a matrix of origins and destinations incorporating the road network and historical data on traffic conditions (Google 2022). We used this approach because traditional distance models such as Euclidean or Manhattan are not accurate when dealing with irregular networks and different travel time models should be used when modeling EMS (Reuter-Oppermann et al. 2017). In addition, new data sources such as geolocated phone data or GPS data can increase the accuracy of urban planning approaches (García-Albertos et al. 2019).

4.2 Parameters and assumptions

Next we list the parameters and assumptions we used in the experimentation.

- In France, the maximum response time for arriving at emergency sites is 30 min in remote areas (Ministère des Solidarités et de la Santé 2015). Therefore, in our case study δ is 30 min.
- An ambulance has a preparation time (between the call arrival and the ambulance departure) of 4 min (Pappinen and Nordquist 2022).
- The on-route emergency eVTOL average airspeed is 3.2 km/min (Holden and Goel 2016).
- For the emergency eVTOL aircraft, we assumed a preparation time between receiving the emergency call and the emergency eVTOL take-off of 6 min.
- The emergency eVTOL aircraft flies in a straight line without considering special air space sectors, such as airports where conflicts with other aircraft can occur.
- For the OS1 and OS2 studied using the methodology of Subsect. 3.1.1, we generated 10,000 clusters of potential emergency sites.
- We did not consider transferring times or admission times to hospitals. We assumed that emergency departments are ready to receive the patients being transported by the emergency eVTOL.

Table 1 Emergency eVTOL aircraft range assumptions, source: (Goyal and Cohen 2022)

Type of eVTOL propulsion	Range (km)		
	Min	Max	Average
Multi rotor	30	50	40
Lift and cruise	50	80	65
Tilt rotor	90	150	120
Tilt duct	110	180	145

- For computing the eVTOL operational service time, we assumed an on-site treatment time of 45.9 min based on the study of Aboueljinane et al. (2014) conducted in the French Val-de-Marne department.

Lastly, to evaluate the impact of the eVTOL range parameter ϕ , we considered four eVTOL propulsion types as seen in Table 1. We determined ϕ as the average between the minimum and maximum range based on the work of Goyal and Cohen (2022). The processing and data analysis of the large amount of data (millions of data points) was performed using the software Python. In addition, the facility location model was run using the software IBM CPLEX 20.1.0. The experiments were run on a PC with Intel Core i7–10870 H, 16 GB RAM and 2.2 GHz of processor. The next section presents the results and analysis of the experimentation.

5 Results and analysis

This section analyzes and compares the behavior of the three OSs under study as follows:

1. Locating the vertiports in existing healthcare facilities, traveling to the patients' location, and transporting them back to the emergency department where the eVTOL departed.
2. Locating the vertiports in existing healthcare facilities, traveling to the patients' location, transporting them to the nearest emergency department, and returning to the vertiport.
3. Locating the vertiports' new location, traveling to the patients' location, transporting them to the nearest emergency department, and returning to the base.

We analyzed the EMS performance measures of (a) coverage, (b) response time, and (c) operational service time. Figure 8 presents the response process as a timeline where the components of the response time and operational service time performance measures can be observed. For those analysis we set the time-savings

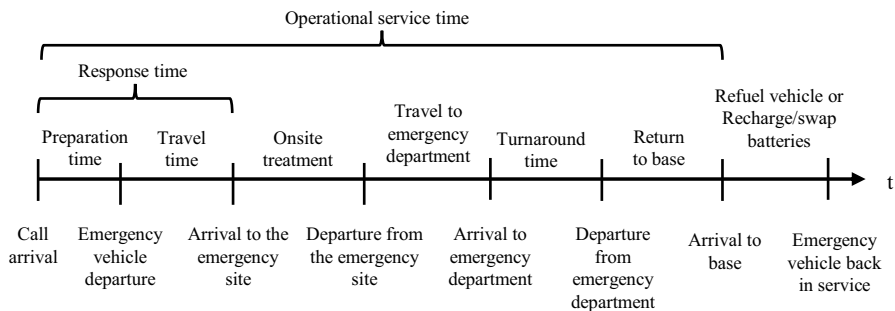


Fig. 8 Emergency response process, adapted from Reuter-Oppermann et al. (2017)

objective parameter λ to 75% (baseline scenario). Next we perform a sensitivity analysis of this parameter and lastly we report data on the approach computational performance.

5.1 Coverage analysis

The first performance measure in our analysis is the coverage. It determines the percentage of sites of the hard-to-reach zones located within range of an emergency eVTOL vehicle. Equation 12 computes the coverage percentage of OS1 and OS2, since the locations were obtained using the facility location model of Subject. 3.1.3.

$$Coverage_1 = \frac{\text{Objective function value of model (Eqs. 6-11)}}{\text{Total of sites of the hard-to-reach zones}} \times 100 \quad (12)$$

For calculating the coverage of OS3, we first determine for each potential emergency site $i \in I$ located in (Lat_i, Lon_i) the nearest opened vertiport $v \in P$ located in (Lat_v, Lon_v) . Next we identify for each site $i \in I$ the nearest healthcare facility with an emergency department $(LatN_i, LonN_i)$. With those locations, we can now calculate if an emergency eVTOL of range $\emptyset$ can perform an emergency mission from the nearest vertiport to the emergency site, transport the patient to the nearest emergency department and return to the vertiport. Equations 13 and 14 calculate the coverage for OS3.

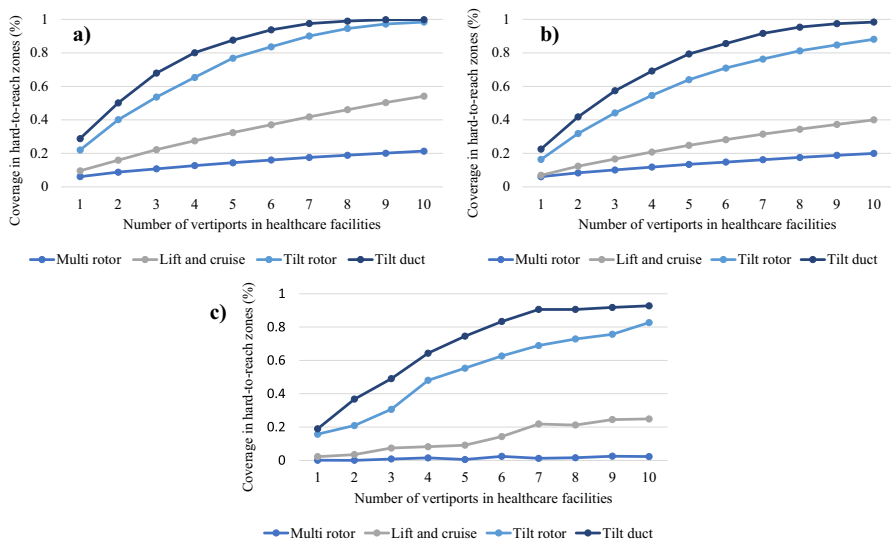


Fig. 9 The percentage of the hard-to-reach zones covered varying the number of vertiports with **a** OS1, **b** OS2, and **c** OS3

$$a_i = \begin{cases} 1, & \text{if } d_{ij}(Lat_i, Lon_i, LatN_v, LonN_v) + d_{ij}(Lat_i, Lon_i, LatN_i, LonN_i) \\ & + d_{ij}(LatN_i, LonN_i, Lat_v, Lon_v) \leq \emptyset \\ 0, & \text{Otherwise} \end{cases} \quad (13)$$

$$Coverage_2 = \frac{\sum_{i \in I} a_i}{\text{Total of hard-to-reach sites}} \times 100 \quad (14)$$

Figure 9 presents the percentage of the sites located in the hard-to-reach zones that are covered using the OS under study varying the number of vertiports. Clearly, the larger the number of vertiports, the higher the coverage in the hard-to-reach zones will be. The results showed an average increase of 5.7%, 4.6%, and 6.6% of coverage in the hard-to-reach zones per additional vertiport in OS1, OS2 and OS3, respectively. In terms of coverage for this case study, locating the vertiports in existing healthcare facilities with emergency departments is the best choice. Besides, the emergency eVTOL of OS1 covered 6.7% and 15.9% more hard-to-reach zones than OS2 and OS3, respectively; and OS2 covered 9.1% more hard-to-reach zones than the OS3. In addition, from Fig. 9, we can conclude that the range of the emergency eVTOL aircraft impacts facility location decisions. By increasing or decreasing the range in any of the OS, the percentage of covered hard-to-reach zones varies significantly. The less competitive eVTOL propulsion types are the multi rotor and lift and cruise. For these aircraft, even with ten vertiports, the percentage of hard-to-reach zones covered does not reach 60% in any of the OSs.

These results are in line with the previous conclusions of Goyal and Cohen (2022). The authors concluded that emergency eVTOLs could not compete with

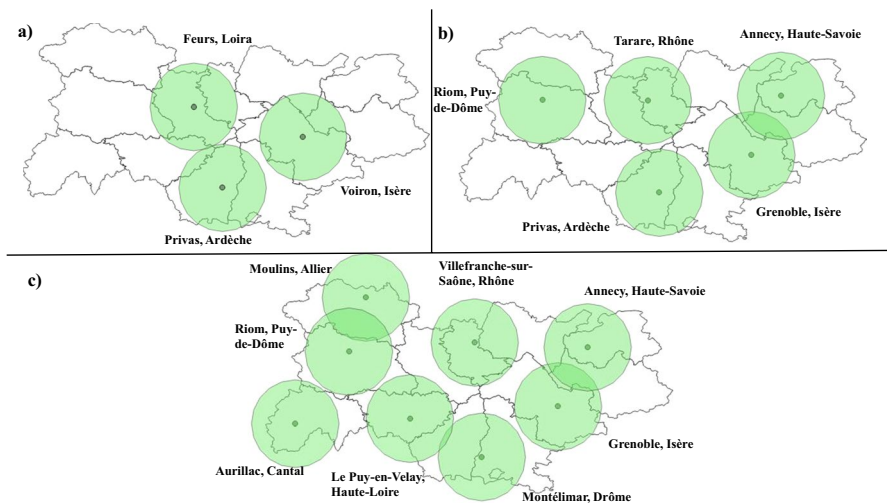


Fig. 10 The geographical location of selected vertiports in the SMUR units for OS1 with an emergency eVTOL autonomy of 120 km for **a** 3, **b** 5, and **c** 8 vertiports

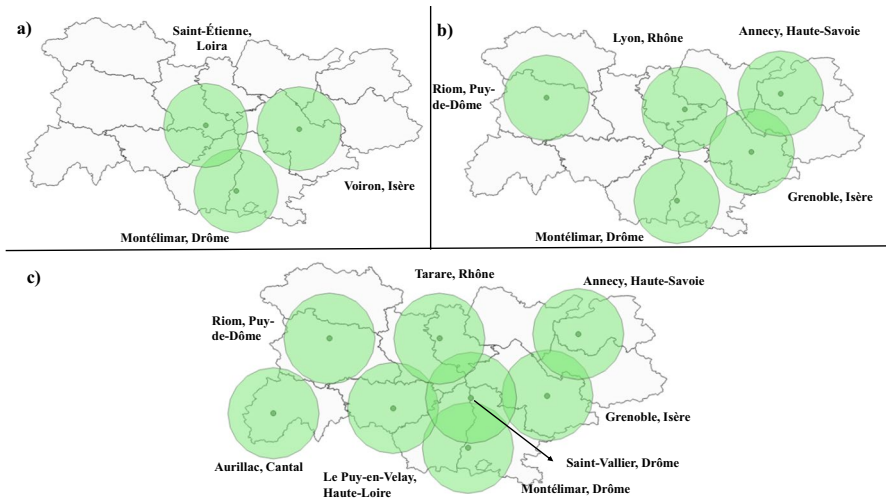


Fig. 11 The geographical location of selected vertiports in the SMUR units for OS2 with an emergency eVTOL autonomy of 120 km for **a** 3, **b** 5, and **c** 8 vertiports

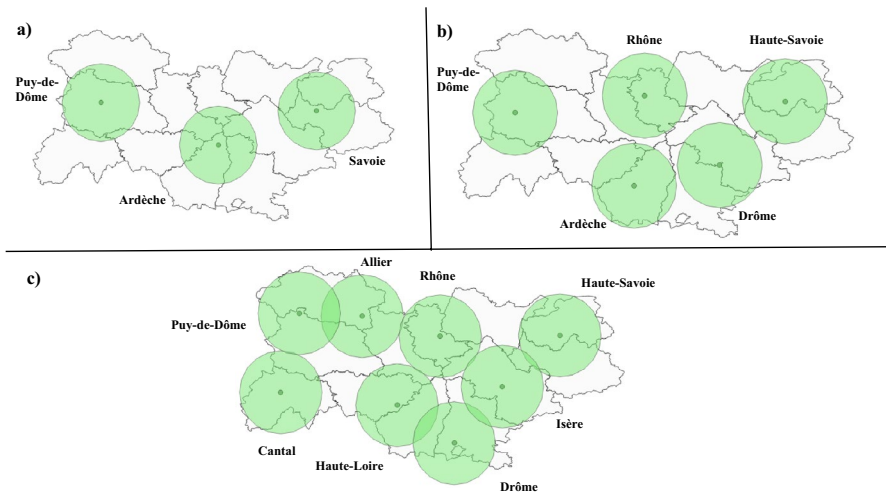


Fig. 12 The geographical location of selected vertiports in new locations for OS3 with an emergency eVTOL autonomy of 120 km for **a** 3, **b** 5, and **c** 8 vertiports

ground ambulances for transportation less than 40 km. The results further suggest that the implementation of AAM technology for EMS in hard-to-reach zones requires the use of aircraft with a minimum operational range of 120 km.

Regarding facility location decisions, Figs. 10, 11 and 12 present maps of the Auvergne-Rhône-Alpes region with locations of the selected vertiports for the tilt rotor aircraft (range of 120 km) in the three OSs. Notably, the selected healthcare

facilities for opening vertiports in OS1 and OS2 are similar. Besides, the maps of Figs. 10, 11 and 12 show how the approaches first establish the vertiports in the region's center, where the sites of hard-to-reach zones with higher number of sites are located. Interestingly, in the departments located in the center of the region, there are multiple healthcare facilities with emergency departments. However, due to traffic conditions, there are still highly populated zones where ambulances do not arrive within 30 min. As more vertiports are allowed to be established in the region, the sites of hard-to-reach zones located in the east and west parts of the region are covered, reaching more than 90% of the hard-to-reach sites of the department with 10 vertiports using the tilt rotor aircraft of 120 km of range.

5.2 Response time analysis

The second performance measure in our analysis is the response time. It corresponds to the time between receiving the emergency call and arrival at the emergency scene (Reuter-Oppermann et al. 2017), as seen in Fig. 8. Since not all the hard-to-reach zones or sites are covered by the emergency eVTOL aircraft due to range constraints, we calculated the response time for the eVTOL uncovered hard-to-reach zones/sites using ground ambulances. Equations 15 and 16 compute the response time indicator for each hard-to-reach zone/site $i \in I$.

$$RT_i = \begin{cases} airt_i, & \text{if the hard-to-reach zone/site } i \text{ is covered by an eVTOL vehicle.} \\ ambt_i, & \text{otherwise.} \end{cases} \quad (15)$$

$$RT = \frac{\sum_{i \in I} RT_i}{\|I\|} \quad (16)$$

Table 2 presents the average response time for the OS under study. Indeed, the response times in hard-to-reach zones improve as the number of vertiports and emergency eVTOL operational range increase. Moreover, OS3 which consists of locating the vertiports inside the hard-to-reach zones, reported better results in terms of response times. However, the improvement is just on average 1.6 and 3.59 min with respect to OS1 and OS2, respectively. The improvement was expected to be more considerable since the vertiports are near the hard-to-reach zones. The reason for this rather contradictory result is that by locating vertiports outside healthcare facilities, the emergency eVTOL aircraft must travel larger distances to the emergency departments and then return to the base. Furthermore, from Table 2 we can conclude that with the eVTOL range of 120–145 km and more than five vertiports in the healthcare network, the system achieves on average the maximum allowed response time of 30 min in France in any of the OSs. In addition, the improvement in the response when establishing additional vertiports is more significant with eVTOL range of 120–145 km.

Table 2 Average EMS system response time for the OS under study

Number of vertiports	Average system response times for OS1 in minutes					Average system response times for OS2 in minutes					Average system response times for OS3 in minutes				
	Aircraft range					Aircraft range					Aircraft range				
	40 km	65 km	120 km	145 km		40 km	65 km	120 km	145 km		40 km	65 km	120 km	145 km	
1	46.813	46.01	42.403	41.19		46.95	46.809	44.58	43.666		45.78	40.985	37.653	38.895	
2	45.896	44.664	38.352	36.642		46.27	46.02	39.66	38.606		41.586	42.54	38.075	32.093	
3	45.357	43.037	33.953	31.863		46.041	44.74	35.86	34.794		43.055	40.261	32.368	29.171	
4	45.047	41.735	31.048	27.656		45.626	43.654	32.903	29.724		40.483	38.516	27.111	22.942	
5	44.437	39.885	26.499	25.634		45.186	42.535	27.905	28.25		41.106	39.718	25.974	22.609	
6	43.883	38.314	25.007	22.144		44.862	41.913	26.978	24.293		41.668	38.151	23.467	19.897	
7	43.242	36.797	21.819	20.584		44.36	40.972	22.93	22.507		40.533	37.145	23.245	17.142	
8	42.972	37.007	19.59	18.829		43.922	40.013	21.698	20.943		42.317	36.185	19.538	17.428	
9	42.471	34.083	18.775	18.628		43.509	39.391	21.696	20.759		40.439	34.569	18.745	16.516	
10	42.161	32.996	17.965	17.683		43.268	38.965	20.587	18.85		41.297	33.414	16.164	15.816	

5.3 Operational service time analysis

This EMS performance indicator measures the average time between an emergency request's arrival and the responding vehicle returning to the base. We analyzed and compared the operational service time for the hard-to-reach zones/sites covered by emergency eVTOL vehicles $i \in I'$ since this study focuses on their implementation. The shorter the operational service time, the sooner the vehicle is available to serve new emergencies. Equations 17 and 18 compute the operational service time indicator for each hard-to-reach zone/site $i \in I$ covered by an emergency eVTOL, where $\|I\|$ denotes the size of the set I .

$$ST_i = RT_i + \text{Onsite} + \frac{d_{ij}(\text{Lat}_i, \text{Lon}_i, \text{Lat}N_i, \text{Lon}N_i) + d_{ij}(\text{Lat}N_i, \text{Lon}N_i, \text{Lat}_v, \text{Lon}_v)}{\text{speed}_{\text{evtol}}} \quad (17)$$

$$ST = \frac{\sum_{i \in I} ST_i}{\|I\|} \quad (18)$$

Table 3 presents the average operational service time for the OS under study for the emergency eVTOL vehicles. As expected, OS1, in which the vertiports are established in healthcare facilities and the patients are transported back, reported the best results. One of the reasons behind this result is that OS1 consists of two flight legs (vertiport to emergency site roundtrip), while OS2 and OS3 consist of three flight legs, as seen in Figs. 2 and 4. On average, OS1 obtained 1.258 and 42.009 min less operational service time than OS2 and OS3, respectively. Interestingly, locating the vertiports outside healthcare facilities requires the eVTOL aircraft to travel more considerable distances delaying the return to the base.

A final thought is that due to the advantages of eVTOL aircraft for aeromedical transportation in terms of range, cooperation strategies between regions can help increase the coverage. As seen in Figs. 10, 11 and 12 the range of some types of eVTOL aircraft exceeds the limits of the region. Therefore, inhabitants of neighboring regions could benefit from this coverage and mechanisms to transport patients between regions should be implemented. In addition, the costs of implementing OS2 and OS3 seem to be higher than OS1 because a higher number of zones must be adapted for eVTOL take-off and landing. Since in OS2 and OS3 the patients are transported to the nearest emergency departments, which can be different to the ones where the vertiport is located, additional vertiports must be established in all healthcare facilities with emergency departments. However, there are still no estimations available of possible costs of vertiports for emergency eVTOL take-off and landing.

5.4 Sensitivity analysis and computational performance

This subsection discusses the effects of variations in the time-savings objective parameter λ in the location of vertiports for EMS. This parameter is one of the two criteria used to determine if a site is classified as a hard-to-reach zone, as seen in

Table 3 Average eVTOL operational service time for the OS under study

Number of vertiports	eVTOL average operational service time for OS1 in min					eVTOL average operational service time for OS2 in min					eVTOL average operational service time for OS3 in min				
	Aircraft range					Aircraft range					Aircraft range				
	40 km	65 km	120 km	145 km		40 km	65 km	120 km	145 km		40 km	65 km	120 km	145 km	
1	60.45	66.907	77.319	80.825		79.241	71.247	79.851	84.918		83.057	99.362	131.657	139.679	
2	60.417	66.904	76.935	82.825		60.271	64.225	78.26	82.225		83.665	100.749	134.981	153.239	
3	60.781	66.399	76.054	81.159		60.382	64.628	77.959	82.482		80.999	97.227	129.507	148.713	
4	60.794	66.505	75.71	80.191		60.382	64.562	75.615	82.359		80.856	96.064	131.346	143.252	
5	60.707	66.176	74.906	79.917		60.493	65.125	76.075	81.15		80.896	98.079	130.826	142.697	
6	60.605	66.068	74.909	77.409		60.729	65.423	76.52	80.515		82.119	95.924	131.202	142.658	
7	60.617	65.953	74.318	77.137		60.72	65.126	76.071	79.301		82.465	97.603	125.48	137.75	
8	60.704	66.064	73.783	75.337		60.611	65.081	75.577	79.166		82.508	97.538	124.822	135.073	
9	60.75	66.03	73.618	75.791		60.669	64.875	75.277	78.955		81.634	96.851	123.015	132.484	
10	60.797	65.768	73.312	73.966		60.716	64.87	74.96	78.54		80.569	96.552	123.137	128.952	

Subsect. 3.1.2. We compared three scenarios with λ equal to 50%, 75%, and 90% denoted as scenarios 1, 2 and 3, respectively. We opted for these particular values based on the number of sites classified as hard-to-reach zones. Preliminary tests showed that low values of λ resulted in almost all clusters being classified as hard-to-reach zones due to the time-savings criterion. Thus, the classification would become meaningless. On the other hand, high values of λ resulted in few hard-to-reach zones and almost all of them were chosen due to the fulfilment of the other criterion (maximum response time not met). Consequently, λ would become meaningless.

As reported above, lower values of λ resulted in more clusters classified as hard-to-reach zones. Out of the 10,000 clusters generated for OS1 and OS2, the scenario 1 we obtained 8,764 hard-to-reach zones, which is higher than the 3,376 hard-to-reach zones obtained in scenario 2. On the other hand, when λ is increased to 90% (scenario 3) the number of hard-to-reach zones decreases to 2,172. This has an impact on the selection of EMS vertiports. As an example, Fig. 13 presents the location of the selected EMS vertiports for OS1, with an eVTOL autonomy of 120 km and eight vertiports for each scenario. Even though the EMS vertiports located in Montelimar, Grenoble, Moulins and Riom were selected in the three scenarios and some vertiports were selected in two scenarios, the remaining selected vertiports were different between scenarios. This behavior was observed in different combinations of OS and emergency eVTOL autonomy.

Furthermore, Tables 5, 6 and 7 of Appendix 2 show the results of the sensitivity analysis. We observe that scenario 1 reported higher coverage values with respect to scenarios 2 and 3 regardless of the emergency eVTOL autonomy or the OS. Besides, scenarios 2 and 3 obtained similar values of coverage. This is because scenario 1 classified highly populated and urban clusters as hard-to-reach zones. Therefore

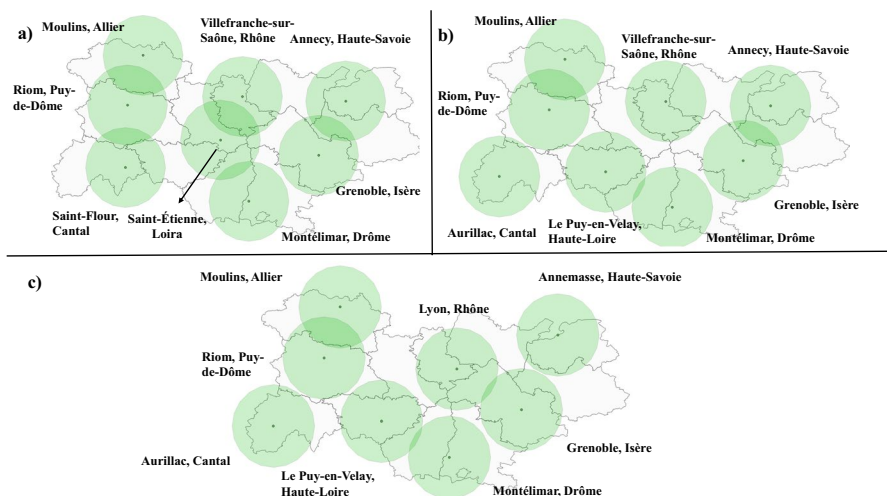


Fig. 13 The geographical location of selected vertiports in the SMUR units for OS1 with an emergency eVTOL autonomy of 120 km for **a** $\lambda = 50\%$, **b** $\lambda = 75\%$, and **c** $\lambda = 90\%$

with fewer vertiports, highly populated zones can be covered increasing this performance indicator. However, this is not desirable because access to EMS would not be improved in zones where the maximum response time is not met or is nearly met. That is why the decision-makers should find the right balance between the number of clusters and the number of hard-to-reach zones by choosing an appropriate value of λ .

In terms of computational complexity of the experiments, the execution time for the proposal tends to increase when the number of vertiports increases.

Table 4 Computational times for the facility location models of OS1 and OS2

λ	Number of vertiports	Computational time in s for OS1				Computational time in s for OS2			
		Aircraft range				Aircraft range			
		40 km	65 km	120 km	145 km	40 km	65 km	120 km	145 km
50%	1	0.67	1.59	13.48	22.86	0.46	0.82	8.40	19.06
	2	0.90	2.02	14.84	18.09	0.63	1.04	9.25	18.38
	3	0.86	2.04	14.06	17.66	0.64	1.05	9.50	23.08
	4	0.87	2.06	14.43	18.25	0.65	1.07	10.18	18.03
	5	0.87	2.12	14.32	33.44	0.62	1.08	10.88	38.24
	6	0.87	2.18	17.26	20.09	0.64	1.17	10.44	21.87
	7	0.89	2.37	15.41	24.36	0.62	1.19	11.54	18.71
	8	0.90	2.27	23.79	43.39	0.63	1.15	17.75	18.99
	9	0.89	2.42	68.11	26.25	0.64	1.14	19.06	20.28
	10	0.93	2.41	19.44	22.86	0.64	1.13	20.98	33.60
75%	1	0.24	0.43	1.48	2.44	0.21	0.42	0.93	2.01
	2	0.29	0.49	1.55	2.73	0.30	0.38	1.15	1.99
	3	0.28	0.50	1.67	2.81	0.30	0.34	1.19	2.11
	4	0.29	0.52	2.55	2.94	0.26	0.34	1.32	1.98
	5	0.30	0.52	1.75	3.37	0.26	0.39	1.32	2.00
	6	0.31	0.66	1.92	6.54	0.27	0.35	1.43	2.57
	7	0.30	0.53	1.84	3.11	0.27	0.34	1.49	2.01
	8	0.31	0.52	1.95	9.07	0.26	0.36	1.30	2.03
	9	0.31	0.53	1.94	2.87	0.26	0.33	1.45	2.15
	10	0.31	0.51	6.17	3.45	0.26	0.34	1.37	4.33
90%	1	0.20	0.28	0.77	1.54	0.15	0.18	0.52	0.75
	2	0.23	0.32	0.81	2.10	0.23	0.21	0.54	1.15
	3	0.18	0.28	0.79	1.85	0.20	0.18	0.61	1.22
	4	0.18	0.29	1.14	1.72	0.19	0.20	0.59	1.16
	5	0.18	0.30	0.82	2.34	0.20	0.20	0.61	1.17
	6	0.17	0.29	0.84	4.07	0.21	0.22	0.65	1.09
	7	0.17	0.30	0.80	1.87	0.24	0.19	0.63	1.06
	8	0.18	0.43	0.88	3.17	0.22	0.18	0.60	1.08
	9	0.17	0.30	1.14	1.50	0.20	0.20	0.59	1.02
	10	0.16	0.44	1.89	1.54	0.21	0.22	1.28	1.79

Further, the computational time increases when the parameter λ decreases as more zones/sites are considered hard-to-reach zones. The facility location instances were solved optimally in short computational times as seen in Table 4. For the k-means algorithms for obtaining the clusters for OS3, on average, the execution time were 734, 543 and 554 s for scenarios 1, 2 and 3, respectively.

6 Conclusions

The latest advancements in eVTOL vehicles indicate that soon this technology will be incorporated into many real-life sectors, including healthcare services. In this work, we studied how the implementation of vertiports can improve access to EMS in hard-to-reach zones. Using data analytics techniques such as clustering and operational research methodologies such as facility location models, we proposed a methodology for locating vertiports to improve the emergency response of healthcare providers. We tested the proposal in a case study in the Auvergne-Rhône-Alpes region, France. Computational results showed that the proposal identifies the hard-to-reach zones where the arrival time of ground ambulances does not meet the required response times or the improvement in the response is more than 75% with AAM technology. Next the results showed that the facility location model can find the optimal location for a certain number of vertiports in EMS facilities, maximizing the coverage of hard-to-reach zones. In this way, AAM technology complements traditional ambulances and can improve the service levels in EMS over a region. In addition, we demonstrated that the operational range of the future eVTOL aircraft used for aeromedical transportation plays an important role in location decisions. Moreover, we corroborated previous findings which indicate that for patient transportation of less than 40 km, it is better to use ground transport. In addition, the main results indicate that, in terms of coverage, locating the vertiports in existing healthcare facilities is the best option. However, the results showed that response times increased compared to locating the vertiports as auxiliary bases outside the healthcare facilities with emergency departments.

Future research should extend our basic problem setting, considering more sophisticated EMS location models such as the Maximum Expected Covering Location Model. In addition, future works can consider uncertainties in travel times, on-site times, or preparation times. As well, different clustering methods, such as the DBSCAN algorithm, can be implemented in the methodology. Another direction for future work is to consider facility location decisions and fleet assignment decisions simultaneously. In addition, future research could address tactical and operational decisions, like the rostering problem of pilots and the medical staff in this type of vehicle.

Appendix 1: Haversine formula

This formula computes the shortest distance between any pair of points considering the surface of a sphere (Inman 1849). If the earth is assumed to be a perfect sphere with a radius $R = 6,371$ km, this formula can effectively be used to calculate the distance d_{ij} between two points $i - j$. The latitude and longitude of point i is represented by (Lat_i, Lon_i) while (Lat_j, Lon_j) denotes the geographical coordinates of point j . Equation 19 presents the haversine formula where the internal spherical angle θ is estimated using Eq. 20.

$$d_{ij}(Lat_i, Lon_i, Lat_j, Lon_j) = Earthradius(R) * \theta(\text{in radians}) \quad (19)$$

$$\theta = 2\sin^{-1} \sqrt{\sin^2 \Delta L + \cos(Lat_i) * \cos(Lat_j) * \sin^2 \Delta G} \quad (20)$$

where:

- ΔL = Half the difference in the latitudes of the two points in radians: $(Lat_j - Lat_i)/2$
- ΔG = Half the difference in the longitudes of the two points in radians: $(Lon_j - Lon_i)/2$

Appendix 2: Results of sensitivity analysis

Table 5, 6 and 7 show the results of the sensitivity analysis.

Table 5 Coverage in target zones for all OS with $\lambda = 50^\circ$

Number of vertiports	Coverage in target zones for OS1					Coverage in target zones for OS2					Coverage in target zones for OS3				
	Aircraft range					Aircraft range					Aircraft range				
	40 km	65 km	120 km	145 km		40 km	65 km	120 km	145 km		40 km	65 km	120 km	145 km	
1	11.45%	16.07%	31.98%	38.59%		8.89%	10.99%	26.91%	37.89%		0.10%	2.31%	21.15%	32.73%	
2	16.35%	24.83%	47.67%	58.69%		12.76%	17.20%	42.46%	56.82%		0.05%	6.21%	24.40%	38.54%	
3	20.90%	32.00%	61.66%	74.94%		16.16%	22.55%	56.09%	69.42%		2.15%	11.52%	34.64%	49.98%	
4	25.42%	38.82%	74.78%	84.80%		19.34%	27.34%	67.69%	80.40%		3.82%	13.22%	48.20%	53.58%	
5	29.13%	44.51%	82.85%	90.51%		22.24%	31.98%	75.46%	87.01%		4.17%	13.31%	51.78%	74.58%	
6	32.72%	50.17%	88.95%	94.65%		25.07%	36.62%	82.39%	92.40%		6.50%	23.47%	70.94%	82.57%	
7	36.09%	54.52%	92.90%	98.03%		27.81%	41.07%	85.92%	96.22%		3.37%	24.23%	72.33%	87.77%	
8	39.22%	58.79%	95.45%	99.17%		30.38%	44.70%	88.68%	98.09%		6.85%	30.42%	77.90%	89.36%	
9	42.02%	62.96%	97.38%	99.68%		32.95%	48.27%	91.43%	98.98%		5.40%	38.98%	83.14%	94.00%	
10	44.52%	66.34%	98.83%	99.80%		35.21%	51.77%	94.14%	99.43%		7.23%	41.39%	88.55%	94.51%	

Table 6 Coverage in target zones for all OS with $\lambda = 75\%$

Number of vertiports	Coverage in target zones for OS1					Coverage in target zones for OS2					Coverage in target zones for OS3				
	Aircraft range					Aircraft range					Aircraft range				
	40 km	65 km	120 km	145 km		40 km	65 km	120 km	145 km		40 km	65 km	120 km	145 km	
1	6.09%	9.60%	21.99%	28.82%		6.08%	6.91%	16.38%	22.57%		0.05%	2.25%	15.71%	18.92%	
2	8.80%	15.96%	40.11%	50.17%		8.36%	12.35%	31.86%	41.77%		0.02%	3.49%	20.88%	36.76%	
3	10.79%	22.23%	53.62%	67.92%		10.11%	16.63%	44.17%	57.39%		0.82%	7.40%	30.62%	49.03%	
4	12.72%	27.49%	65.34%	80.10%		11.81%	20.80%	54.61%	69.16%		1.47%	8.18%	48.05%	64.34%	
5	14.39%	32.40%	76.85%	87.62%		13.43%	24.81%	64.00%	79.36%		0.49%	9.13%	55.30%	74.52%	
6	16.02%	37.07%	83.62%	93.76%		14.82%	28.19%	70.98%	85.60%		2.41%	14.29%	62.63%	83.35%	
7	17.59%	41.80%	90.08%	97.55%		16.21%	31.49%	76.31%	91.66%		1.23%	21.82%	68.93%	90.60%	
8	18.88%	46.08%	94.53%	99.00%		17.59%	34.45%	81.26%	95.39%		1.55%	21.23%	72.83%	90.57%	
9	20.13%	50.34%	97.24%	99.81%		18.83%	37.29%	84.74%	97.46%		2.49%	24.47%	75.65%	91.79%	
10	21.36%	54.17%	98.39%	99.88%		19.98%	40.02%	88.14%	98.41%		2.26%	24.85%	82.67%	92.76%	

Table 7 Coverage in target zones for all OS with $\lambda = 90^\circ$

Number of vertiports	Coverage in target zones for OS1					Coverage in target zones for OS2					Coverage in target zones for OS3				
	Aircraft range					Aircraft range					Aircraft range				
	40 km	65 km	120 km	145 km		40 km	65 km	120 km	145 km		40 km	65 km	120 km	145 km	
1	6.34%	10.07%	23.11%	30.45%		6.34%	7.21%	17.64%	23.51%		0.04%	3.12%	14.28%	19.45%	
2	8.92%	17.27%	42.36%	51.17%		8.44%	13.23%	34.36%	43.08%		0.06%	3.98%	21.65%	29.15%	
3	11.08%	23.96%	56.30%	69.18%		10.31%	17.83%	45.30%	57.78%		0.95%	6.17%	28.59%	45.56%	
4	13.18%	29.29%	67.00%	81.28%		12.11%	21.70%	55.51%	70.24%		1.32%	8.21%	50.15%	65.74%	
5	15.10%	33.97%	77.55%	88.46%		13.81%	25.48%	65.30%	80.23%		1.39%	9.65%	56.37%	71.15%	
6	16.58%	38.61%	84.38%	94.25%		15.44%	29.15%	71.83%	86.45%		2.12%	12.31%	59.23%	80.87%	
7	17.91%	43.09%	90.41%	97.69%		16.67%	32.16%	76.90%	91.52%		2.18%	18.64%	64.39%	88.31%	
8	19.12%	47.30%	94.65%	99.12%		17.88%	35.18%	81.97%	94.90%		1.98%	18.45%	66.30%	91.67%	
9	20.31%	51.54%	96.94%	99.77%		19.01%	37.90%	85.53%	97.25%		2.78%	22.37%	72.98%	92.13%	
10	21.46%	54.87%	98.24%	99.84%		20.12%	40.57%	88.52%	98.34%		3.02%	23.50%	80.15%	94.15%	

Acknowledgements This work was made under the project grant INGPhD-44-2021 awarded by the School of Engineering at Universidad de La Sabana, and by an Eiffel Excellence Scholarship from the French Ministry for Europe and Foreign Affairs awarded to the first author.

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Funding Open Access funding provided by Colombia Consortium. Eiffel Excellence Scholarship from the French Ministry for Europe and Foreign Affairs awarded to the first author and Project grant ING-PhD-44-2021 by School of Engineering at Universidad de La Sabana.

Data availability The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Code availability The code generated during the current study is available from the corresponding author on reasonable request.

Declarations

Conflict of interest All authors declare that they have no conflicts of interest.

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